

Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. LCLS Parameters 20 pts.

Consider the parameters of LCLS:

Parameters	Values
Beam energy	14.35 GeV
Peak current	3500 A
Pulse length	230 fs
FEL parameter	5×10^{-4}
RMS angular divergence	$1.7 \mu\text{rad}$
Undulator wavelength	3 cm
Undulator parameters K	3.7

Calculate the following:

- 1) Radiation wavelength, λ_r .
- 2) Bunch charge of each electron bunch.
- 3) Approximate number of electrons in one micro-bunch.
- 4) 1-D gain length, L_{1D} .
- 5) Approximate undulator saturation length.
- 6) Approximate peak radiation power.
- 7) The width of the frequency spectrum at saturation.

Problem 2. Pillbox Cavity 30 pts.

Consider a hollow copper pillbox cavity of radius $a = 75/2$ cm and axial length $L = 50$ cm. Assume the interior volume of the cavity is under vacuum.

- a) 2 pts: What is the resonant frequency of the fundamental TM_{010} mode?
- b) 6 pts: Calculate the transit time factor for a particle with $\beta = 0.85$.
- c) 5 pts: Explain how the cavity operating at the fundamental frequency might be modified to increase the acceleration efficiency (larger T for given β found in part c). Qualitative only.

- d) 15 pts: For the cavity operating at the fundamental frequency, assume an RF voltage $V_0 = E_0 = 500$ kV and assume the cavity is made of copper with conductivity of $\sigma = 5.9 \times 10^7$ S/m, calculate:

U stored EM energy

R_{surf} RF surface resistance

P_{cy} average power lost over RF period

Q_0 Intrinsic Quality factor

R_s shunt impedance

ΔE_{kin} Particle kinetic energy gain

Use formulas derived in the class notes. **Include units in your answers for all physical quantities.** Note: $J_1(2.405) = 0.519$.

Problem 3. Synchrotron Energy Acceptance 35 pts.

Consider the continuous approximation RF focusing/acceleration equations derived in class for a ring:

$$\frac{d}{dn}\phi = -\frac{2\pi h\eta_s}{\gamma_s\beta_s^2} \frac{\Delta W}{mc^2}$$

$$\frac{d}{dn}\Delta W = qV_{rf}(\sin\phi - \sin\phi_s)$$

with $\eta_s = \frac{1}{\gamma_s^2} - \frac{1}{\gamma_{tr}^2}$ the usual slip factor. Assume that the coefficients (i.e., β_s , η_s , and V_{rf}) remain approximately constant lap-by-lap in the evolution.

- a) 10 pts: Show for a particle evolving in the bucket that

$$\frac{1}{2} \left(\frac{d\phi}{dn} \right)^2 - \frac{2\pi h\eta_s}{\gamma_s\beta_s^2} \frac{qV_{rf}}{mc^2} (\cos\phi + \phi \sin\phi_s) = \text{const}$$

Hint: Combine to make one equation. Multiply by $\frac{d\phi}{dn}$. Then integrate with respect to dn .

- b) 5 pts: Show that the result in a) can be expressed as:

$$\Delta W^2 - \frac{\beta_s^2 \gamma_s^2 mc^2 q V_{rf}}{\pi h \eta_s} (\cos\phi + \phi \sin\phi_s) = \text{const}$$

- c) 10 pts:

Choose ϕ_s for maximum acceptance (corresponding to no acceleration) below transition and show that for this case that the separatrix equation is:

$$\Delta W^2 - \frac{2\gamma_s\beta_s^2 mc^2 q V_{rf}}{\pi h \eta_s} \cos^2\left(\frac{\phi}{2}\right) = 0$$

Hint: It is easy to identify the constant at the x -points at $\phi = \pm\pi$.

- d) 10 pts: Apply the result in part c) to estimate the energy acceptance $\Delta W/W$ of a proton synchrotron with:

$h = 80$ harmonic number
 $\eta_s = 0.05$ slip factor
 $V_{rf} = 1$ MV rf Voltage
 $W = 200$ MeV Kinetic Energy

Assume incoming particles have the correct synchronous phase and that the machine is operated for maximum acceptance.